



Installing and setting up Windows 11 environment

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Purpose

This lab documents the process of setting up a NVM drive with Windows 11 and preparing it for future use in Cisco Classroom Settings.

Background

Operating systems form the bridge between computer hardware and user applications, providing the environment in which all modern computing tasks are performed. Windows 11 is the eleventh major version of the Microsoft Windows operating system. Its tenure and wide range of compatibility make it the ideal operating system for most scenarios such as office work, schoolwork, and other applications.

History of Microsoft windows:

Microsoft windows has been the dominant operating system for personal computers for over three decades. Its history reflects the evolution of modern computing, from simple graphical environments to advanced, secure, and interconnected platforms.

Windows 1.0 (1985): The first version of Windows was not a standalone operating system but rather a graphical shell that ran on top of MS-DOS. It introduced features such as simple windowed interfaces, mouse support, and basic applications like paint notepad.

VB

Windows 95 (1995): Introduced the Start Menu, taskbar, and plug-and-play hardware support.

Windows 98 and ME (1998–2000): These releases focused on home computing, adding USB support, multimedia capabilities, and internet integration. However, they were also criticized for instability, especially Windows ME.

Windows NT and 2000: Designed for business and enterprise environments, these operating systems prioritized stability, security, and multi-user capabilities. They laid the foundation for Microsoft's future operating system architecture.

Windows XP (2001): One of the most successful versions of Windows, XP combined the consumer-friendly interface of Windows 98 with the stability of the NT kernel. It remained widely used for over a decade and set the standard for user experience.

Windows Vista (2007): Introduced a new visual style called Aero and security features like User Account Control (UAC).

Windows 7 (2009): praised for its speed, reliability, and improved interface.

Windows 8 and 8.1 (2012–2013): introduced a tile-based Start Screen optimized for touch devices. However, the removal of the traditional Start Menu was unpopular, leading to mixed adoption.



Windows 10 (2015): Microsoft repositioned Windows as a “service,” delivering continuous updates instead of new major versions. Windows 10 reintroduced the Start Menu and emphasized cloud integration, security, and support for a wide range of devices.

Windows 11 (2021–present): The latest generation of Windows introduced a redesigned interface with centered taskbar icons, rounded corners, Snap Layouts for multitasking, and tighter integration with Microsoft Teams. It also placed strong emphasis on hardware-based security, requiring TPM 2.0 and modern processors.

Microsoft Office 365 is installed to provide productivity tools such as Word, Excel, and PowerPoint, supporting both academic assignments and professional tasks

NVIDIA Graphics Driver is installed to ensure the GPU runs at full performance, providing advanced rendering, and stability for multimedia workloads

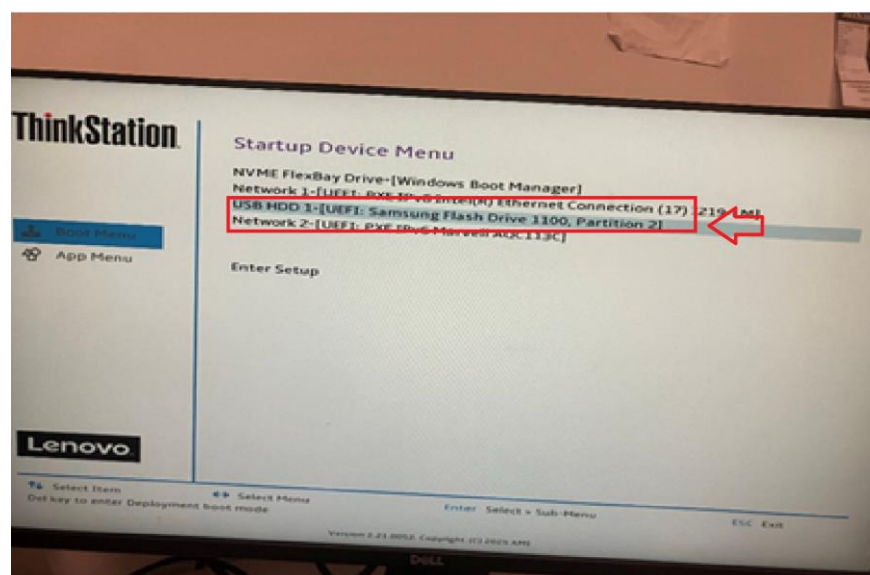
PuTTY is an open-source terminal emulator software developed by Simon Tatham. It has long been used to emulate serial connections and as an SSH and telnet Client in networking solutions.

Wireshark is a packet analysis tool used to monitor, capture, and troubleshoot network traffic. While it has many legitimate applications, Wireshark can also be used for malicious purposes, as captured traffic can give information about potential vulnerabilities in network infrastructure.

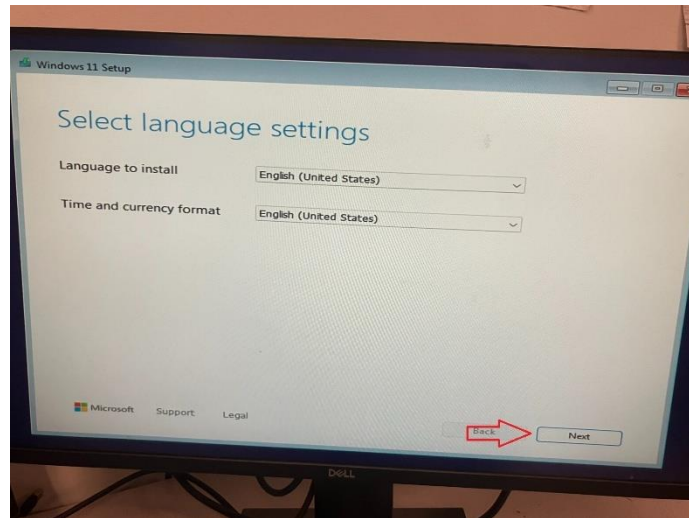
Lab Commands

Windows 11 installation

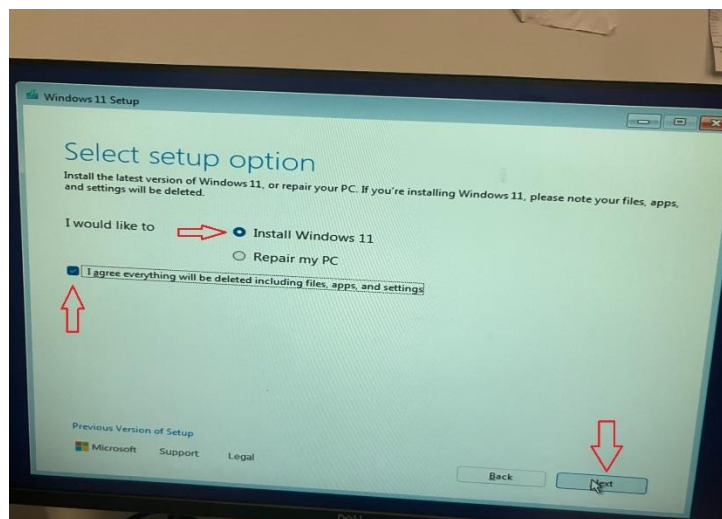
- Create a bootable media drive from the Windows 11 media creation tool (<https://www.microsoft.com/software-download/windows11>) (in this case, we were provided a premade USB drive).
- With the USB inserted, turn the PC on and enter the bios by clicking the F12 button on your keyboard.
- When in the Bios, Boot from the USB drive



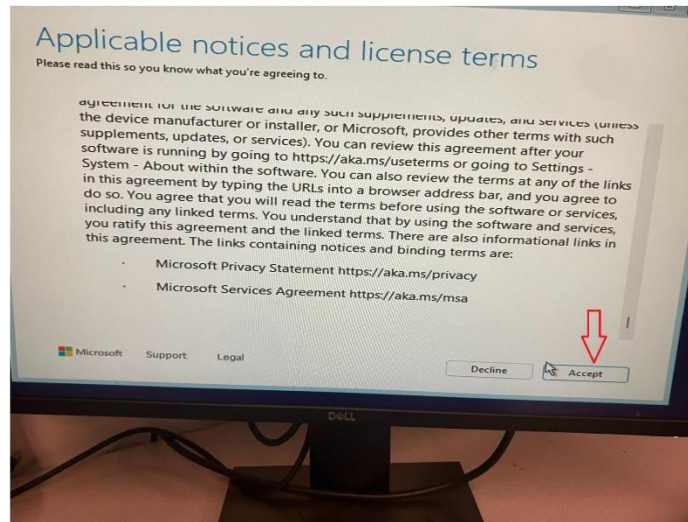
- Select the language to install and the time and currency format that you desire



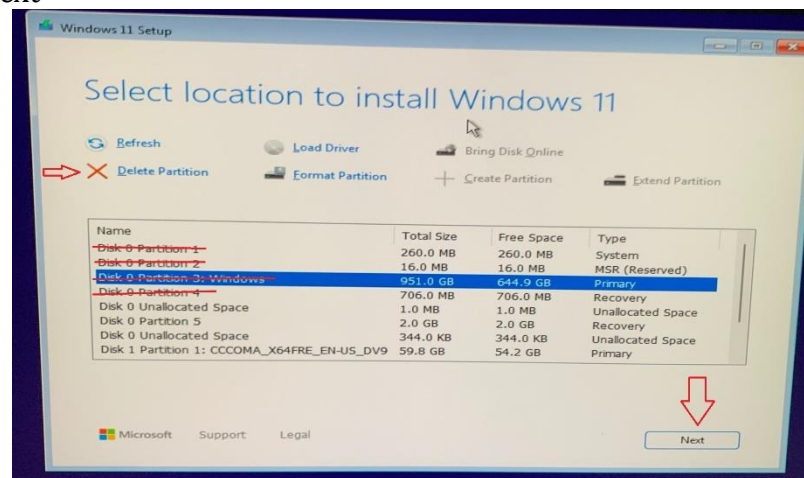
- Select the install Windows 11 option and select the box that says you agree that everything will be deleted including files, apps, and settings.



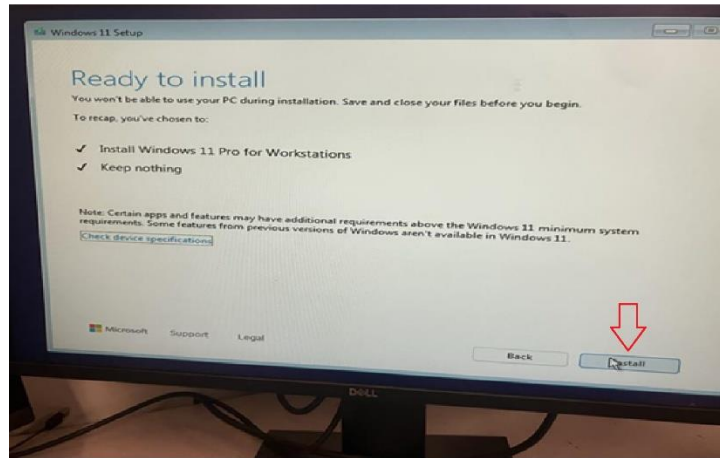
- Accept all the applicable notices and license terms



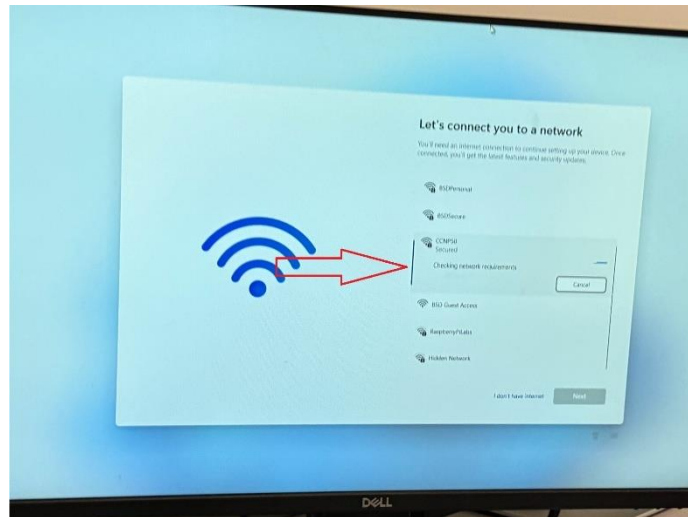
- Delete all the disk partitions for Disk 0 except the partition for unallocated space and click "Next"



- Click on "Install" to install windows 11



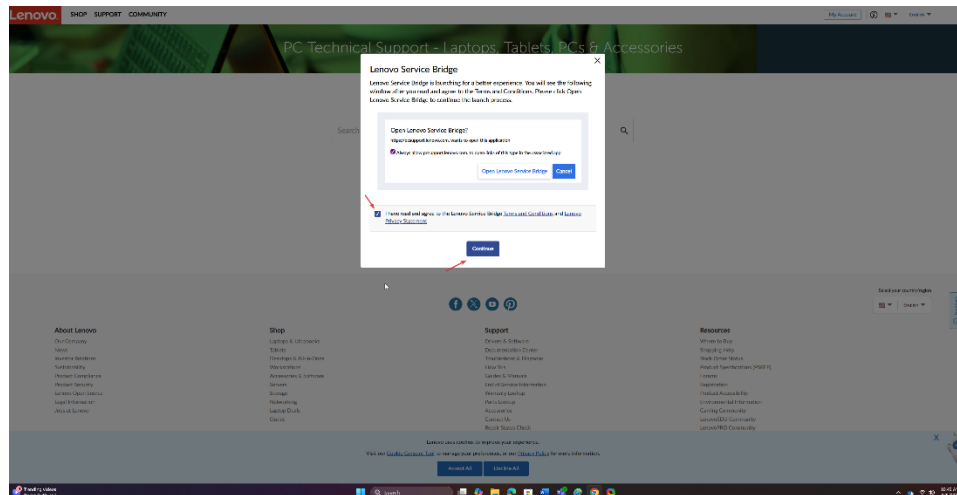
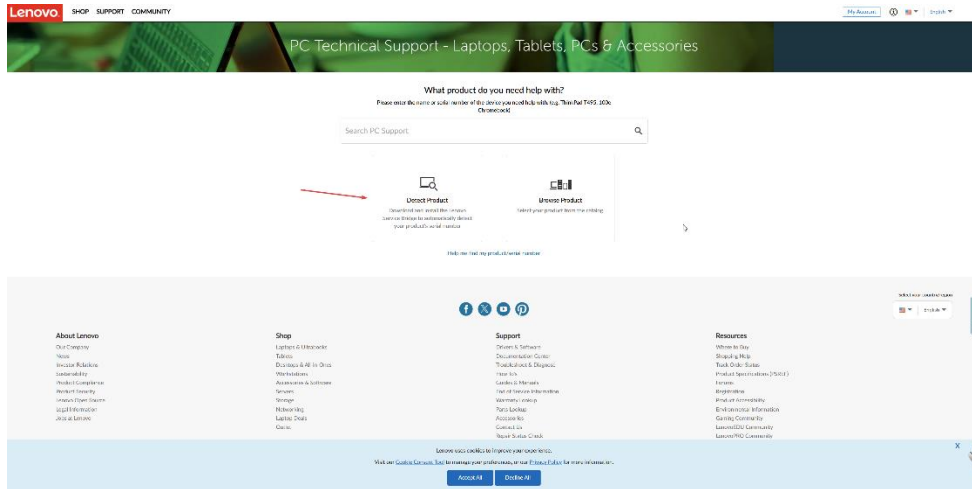
- Connect to Your wifi network



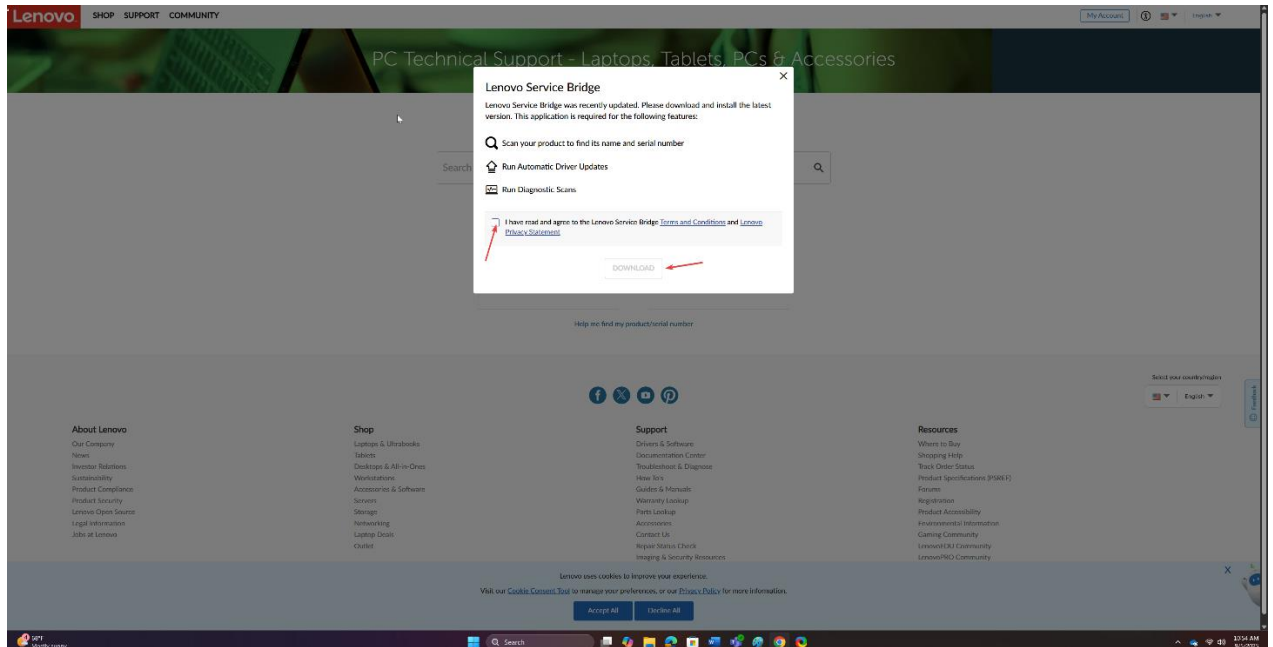
- Be agreeable through the rest of the install, and you will eventually arrive at the desktop

Drivers:

- In a browser, Navigate to <https://pcsupport.lenovo.com/us/en/>
- Under PC, click Detect Product and allow Lenovo Service bridge to run. If it returns an error, click the link at the bottom of the pop-up to download Lenovo Service Bridge.



- If it doesn't detect your device click on the popup to download Lenovo service bridge



- Once Lenovo Service bridge detects the workstation's model and specifications, click Drivers and Software and then click Automatic Update
- Lenovo Service Bridge will update any necessary drivers.
- At this point, restart your computer by pressing the start key, pressing the power button, and pressing Restart.

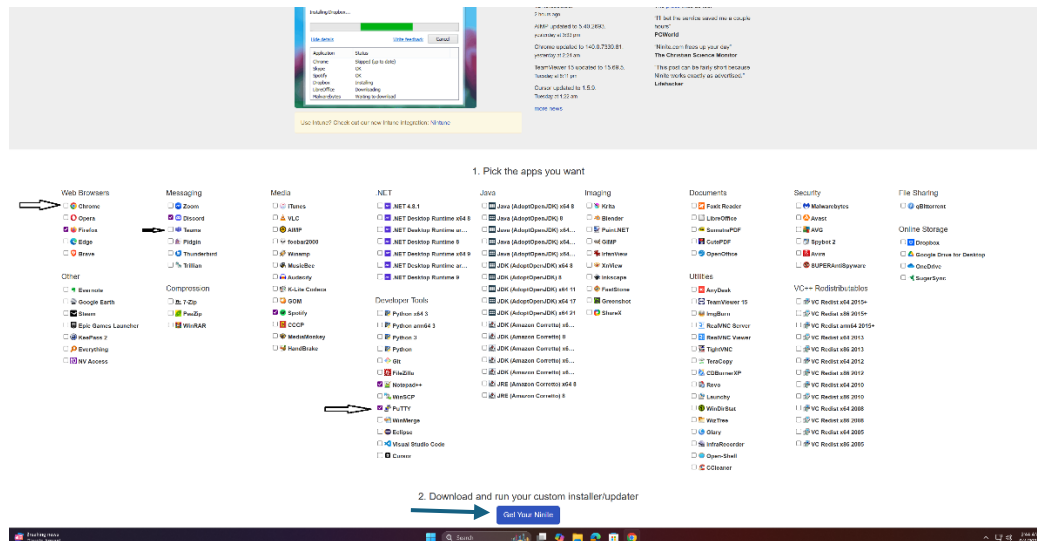
Software:

PuTTY, Google chrome, Teams

- In a internet brower navigate to <https://ninite.com/>



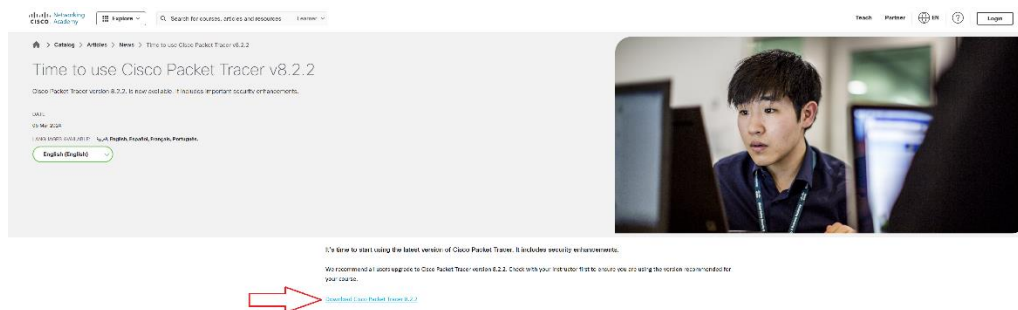
- Scroll down and click on as many apps to download at once in bulk. Particularly select Putty, Chrome, and Teams.



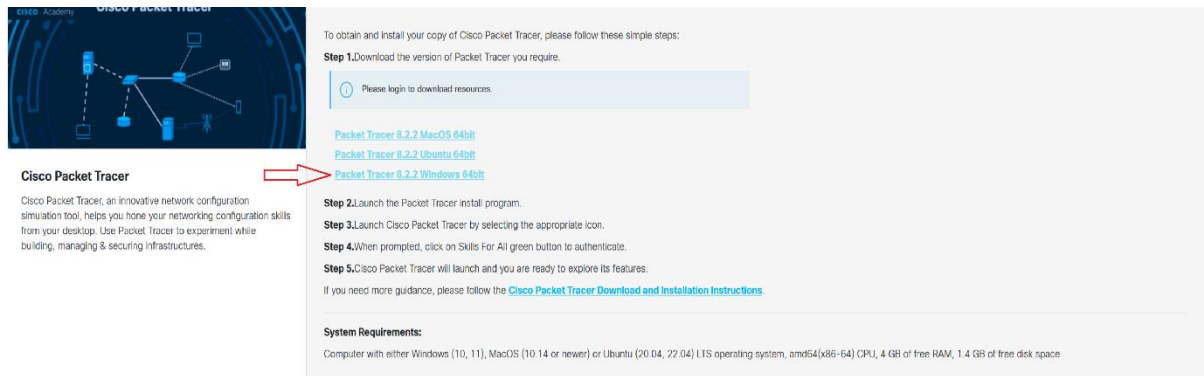
- Click on get your ninte and this software will be installed on your PC

Packet Tracer

- In an Internet Browser navigate to <https://www.netacad.com/articles/news/download-cisco-packet-tracer>
- Click on the blue “Download Cisco Packet Tracer 8.7.2” url:



- Scroll down and download the Packet tracer for the Operating System that you are on (Windows 11 in this case)



To obtain and install your copy of Cisco Packet Tracer, please follow these simple steps:

Step 1.Download the version of Packet Tracer you require.

Please login to download resources.

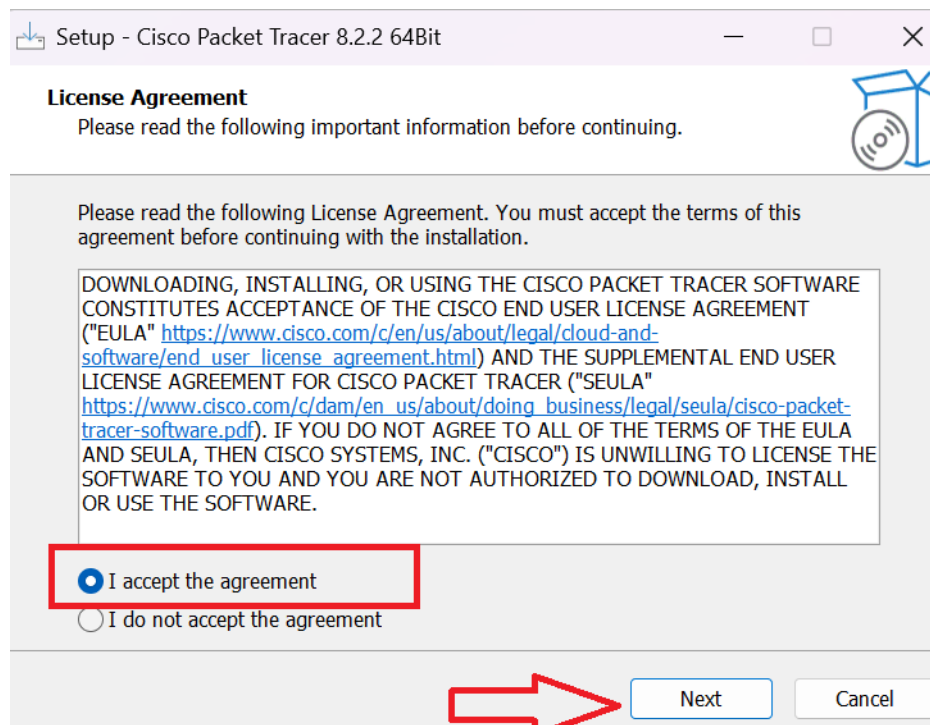
[Packet Tracer 8.2.2 MacOS 64bit](#)
[Packet Tracer 8.2.2 Ubuntu 64bit](#)
[Packet Tracer 8.2.2 Windows 64bit](#)

Step 2.Launch the Packet Tracer install program.
Step 3.Launch Cisco Packet Tracer by selecting the appropriate icon.
Step 4.When prompted, click on Skills For All green button to authenticate.
Step 5.Cisco Packet Tracer will launch and you are ready to explore its features.

If you need more guidance, please follow the [Cisco Packet Tracer Download and Installation Instructions](#).

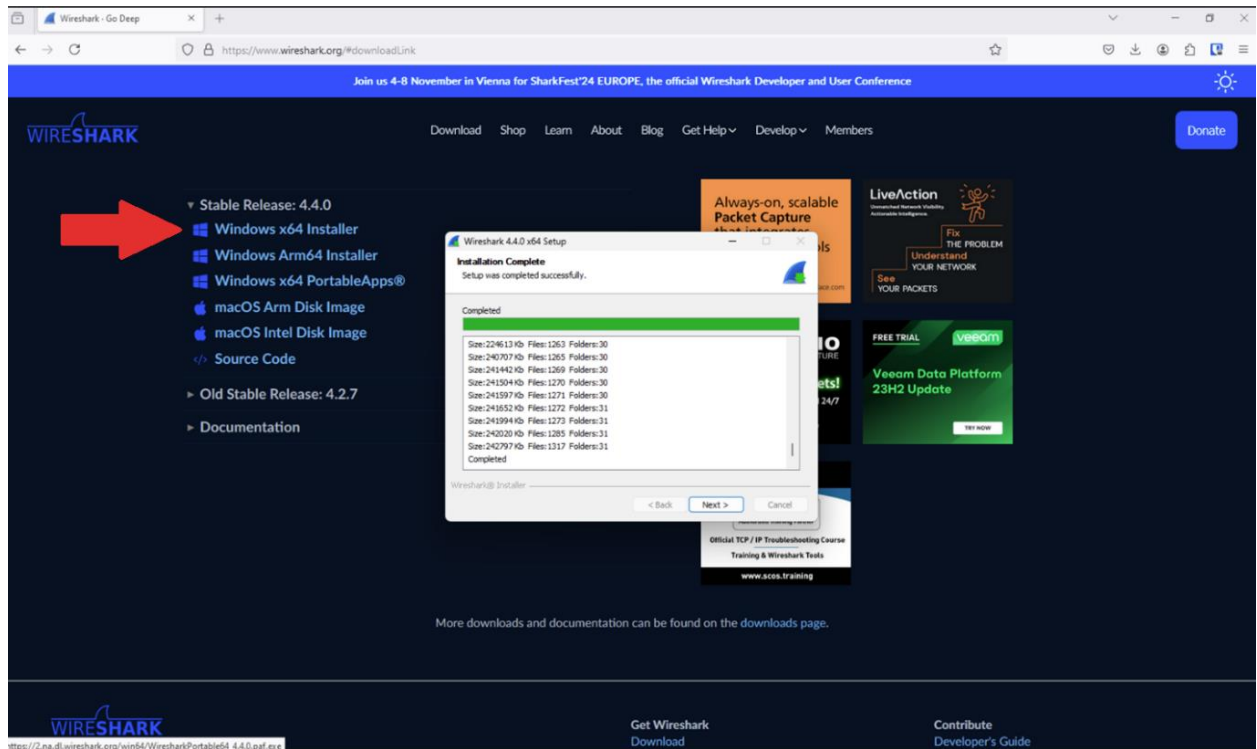
System Requirements:
Computer with either Windows (10, 11), MacOS (10.14 or newer) or Ubuntu (20.04, 22.04) LTS operating system, amd64(x86-64) CPU, 4 GB of free RAM, 1.4 GB of free disk space

- Once Downloaded, open the installer, agree to the terms, and packet tracer will be downloaded.



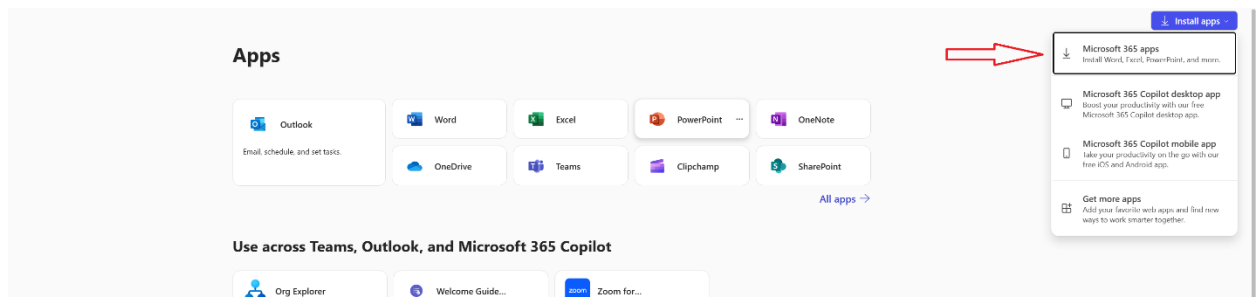
Wireshark

- Navigate to <https://www.wireshark.org>, click Download, then click x64 Windows Installer and run the resulting .exe file. Be agreeable through the installation.

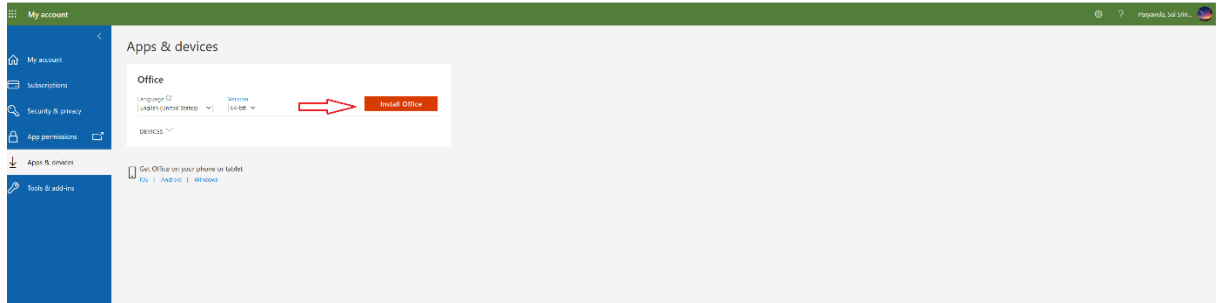


Microsoft Office 365

- On a Web browser go to: <https://m365.cloud.microsoft/apps/?auth=2>
- Click on the Install apps icon and select the Microsoft 365 apps option



- Select the preferred language and click on install office. Be agreeable through the installation and will have all the Office 365 apps



Problems

One of the first issues was an issue with NVMe drives corrupting. I had my first drive corrupt. This occurred because the drive was ejected from the NVMe expansion bay slot while the workstation was running. This issue can be avoided by being vigilant and making sure that the workstation is completely powered off before removing/inserting a drive.

Second, when installing the drivers the Lenovo website had a difficult time recognizing the device of the PC, and it would not successfully download the Lenovo service bridge to get the necessary driver. It seemed to be because of a network error in the lab or something else.

Third, I had problems with slow internet. Programs like the Lenovo service bridge, NVIDIA app, putty, and Wireshark need pretty fast internet to download and run. The internet service I was working with was pretty slow, which made the whole process of downloading software slow as well. To mitigate this, I used my own personal hotspot from my phone for most of the downloading.

Conclusion

All in all, the installation process of Windows 11 and the various software required went smoothly, with only a few hangups in the process. Microsoft's intrusive interface design made installation slightly more difficult, but still very achievable by the average Cisco student.

